Design and Validation of an Early Warning System for Anxiety Crises using Deep Learning on Single-Lead ECG and Computer Vision: Project CardioCalm AI

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***Abstract*—Anxiety is a growing public health issue, exacerbated by the COVID-19 pandemic, significantly affecting the young adult population in Peru. Current diagnostic methods, based on self-reports, lack objectivity and continuity. This project presents CardioCalm AI, an end-to-end system for stress and anxiety detection using physiological signals. We trained and benchmarked 7 Machine Learning and Deep Learning architectures using the multimodal WESAD dataset. Results demonstrate that the InceptionTime architecture, fed solely with raw Electrocardiogram (ECG) signals, achieves an F1-Score of 0.958 in stress detection, outperforming traditional configurations that rely on multiple sensors (EDA, Temperature). The solution was deployed on an ESP32-based wearable prototype and a Progressive Web App, complemented by a computer vision module for secondary emotional validation.**

***Index Terms*—Anxiety, Deep Learning, ECG, InceptionTime, WESAD, Affective Computing, Wearable.**

I. Introduction

Anxiety constitutes one of the most critical challenges for global mental health in the 21st century. According to the World Health Organization (WHO), anxiety disorders affect over 300 million people worldwide [1]. In Peru, data from EsSalud indicates that in 2024, over 182,000 cases of anxiety disorders were diagnosed [2]. This issue has intensified particularly among young adults (18–29 years old), where the transition to working life acts as a catalyst for chronic stress.

From a physiological perspective, anxiety is a systemic response governed by the Autonomic Nervous System (ANS). Upon a stressor, the sympathetic system triggers measurable changes such as increased heart rate and sweating [3]. Historically, Heart Rate Variability (HRV) has been established as the gold standard for assessing autonomic balance [4]. This project leverages Artificial Intelligence to transform these electrical signals into preventive alerts.

II. Problem Statement

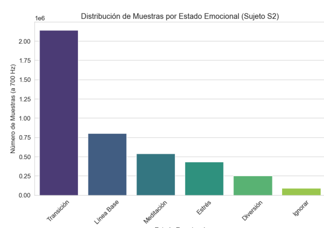
Detecting Generalized Anxiety Disorder (GAD) faces significant barriers: subjectivity in questionnaires (e.g., GAD-7), intermittency in clinical evaluations, and hardware complexity in current monitoring systems requiring multiple sensors.

The central hypothesis of this study is that a single ECG lead, processed via Deep Learning, is sufficient to detect anxiety crises with clinical precision, eliminating the need for auxiliary sensors like Electrodermal Activity (EDA), which are cumbersome and environment-dependent.

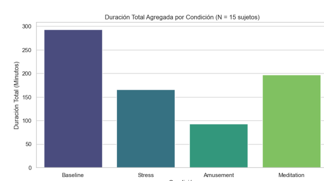
III. Methodology

A. Dataset and Exploratory Data Analysis (EDA)

The WESAD dataset [5] was selected due to its validated stress induction protocol (TSST) involving 15 young subjects. The duration analysis (Fig. 1) revealed a class imbalance (Baseline > Stress), necessitating the use of F1-Score as the primary metric.



(a) Sample distribution per subject.



(b) Class imbalance (Baseline vs Stress).

Figure 1: WESAD Dataset Analysis.

B. Ground Truth Validation

To ensure models learn real stress patterns, psychological self-reports were analyzed. The STAI (State-Trait Anxiety Inventory) scores showed a statistically significant increase during the stress condition compared to baseline, confirming the effectiveness of the induction protocol. Similarly, PANAS questionnaires reflected a spike in negative affect, validating the ground truth labels used for training.

C. Signal Inspection

Visual analysis of raw signals (Fig. 2) showed evident morphological changes in the ECG during stress, characterized by R–R interval reduction (tachycardia) and amplitude alterations. Temperature sensors were evaluated but discarded due to their slow response time to acute events.

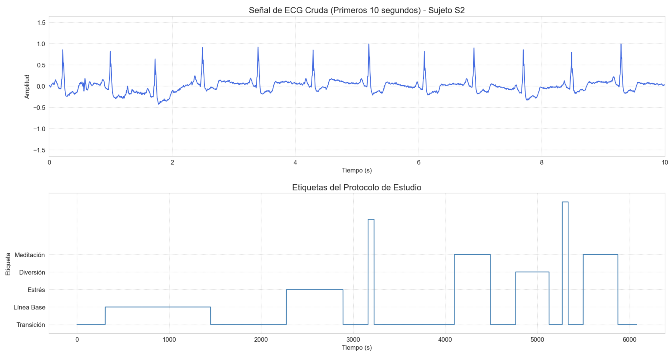


Figure 2: Raw ECG signal visualization. Note the frequency change in the labeled stress zones.

IV. Modeling Strategies

A. Classical Machine Learning (Feature Engineering)

We used *NeuroKit2* [6] to extract 188 features from the time and frequency domains (HRV). Biological validation confirmed that RMSSD drops significantly during stress, indicating vagal inhibition. Four models were trained using these features: **KNN, SVM, Random Forest, and XGBoost**.

B. Deep Learning

We experimented with advanced architectures for raw time-series processing to bypass manual feature extraction. We implemented a Multilayer Perceptron (MLP) on features, a hybrid **CNN-LSTM** network, and the **InceptionTime** model [7]. InceptionTime applies parallel convolutions with different kernel sizes, allowing the network to capture both short-term (QRS complex) and long-term (rhythm) patterns simultaneously.

V. Results and Discussion

A. Model Benchmark

A total of 7 models were evaluated. Fig. 3 illustrates that **XGBoost** and **InceptionTime** achieved the best global performance.

Table I: Performance Comparison (Top Models)

Model	Acc	F1-Macro	Recall (S)	F1 (S)
XGBoost	**0.947**	**0.930**	**0.978**	**0.958**
InceptionTime (Raw)	0.933	0.900	**0.978**	**0.958**
Random Forest	0.900	0.844	0.957	0.957
MLP (Features)	0.928	0.907	0.957	0.957
SVM	0.881	0.826	0.978	0.920

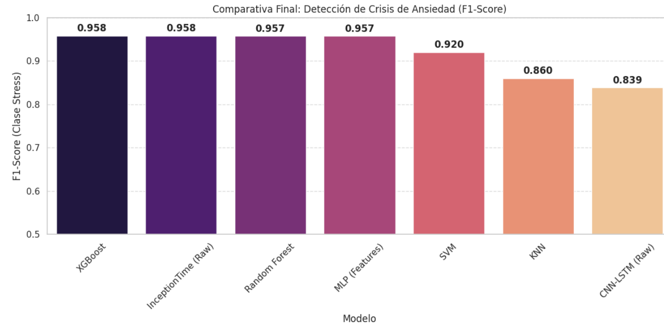
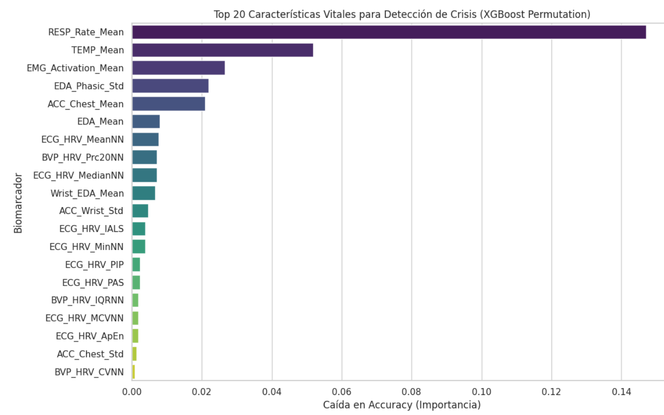


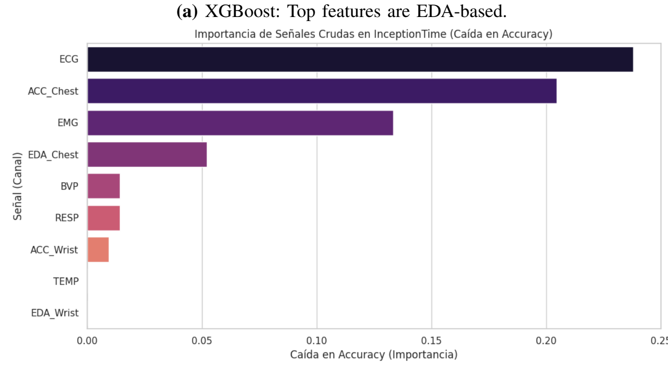
Figure 3: Final Model Ranking (F1-Score for Stress Class). XGBoost and InceptionTime lead the performance.

B. Interpretability and Sensor Ablation

Despite similar scores, the models learn differently. The permutation importance analysis (Fig. 4) reveals a critical finding: **XGBoost relies heavily on EDA features**, while **InceptionTime prioritizes the ECG signal**.



(a) XGBoost: Top features are EDA-based.



(b) InceptionTime: Top channels are ECG and ACC (Chest).

Figure 4: Permutation Importance Analysis. Deep Learning successfully extracts morphological patterns from the ECG, whereas ML depends on the simpler EDA statistics.

This dependency creates a vulnerability for XGBoost in single-sensor scenarios. As shown in Table II, when EDA is removed (”ECG Only” scenario), XGBoost’s performance drops to 0.851, while **InceptionTime maintains a robust 0.979**.

Table II: Ablation Study: F1-Score (Stress) by Sensor Configuration

Scenario	**XGBoost (Features)**	**Inception (Raw)**
Full Model	0.969	0.959
Chest All	0.978	0.989
ECG Only (Target)	*0.851*	**0.979**
Wrist (Wearable)	0.978	0.808

VI. System Implementation

A. Architecture and Hardware

An *End-to-End* solution was developed, comprising a wearable based on ESP32-S3 and the AD8232 sensor (Lead II). The signal is transmitted via BLE to a PWA and processed in Google Cloud Run. The prototype resolves ADC conflicts through pin remapping and displays local data on an OLED screen. (Visuals in Appendix A).

B. User Interface and Computer Vision

The Web App enables Bluetooth connection (Web BLE) and real-time ECG visualization. As an auxiliary validation mechanism, a computer vision module was integrated for facial emotion recognition. This module builds upon the pre-trained convolutional architecture released as part of the MIT representation learning challenge [8], allowing correlation between physiological responses and facial affective expression.

VII. Conclusions

This study validates that a single ECG signal, processed with InceptionTime, is sufficient to detect anxiety crises with an F1-Score of 0.958, effectively eliminating the need for expensive sensors like EDA. The successful implementation on an ESP32 microcontroller demonstrates the technical and economic feasibility of producing low-cost medical devices to address the mental health crisis in Peru.

Project Links

- **Repository:** GitHub GRUPO-06-ISB
- **Web App:** CardioCalm App
- **Demo Creds:** prueba@cardiocalm.com / 123456

References

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Appendix A: Implementation Details

This appendix illustrates the complete system architecture, hardware design, real-time interface, historical logging screen, and facial emotion recognition module used in CardioCalm AI.

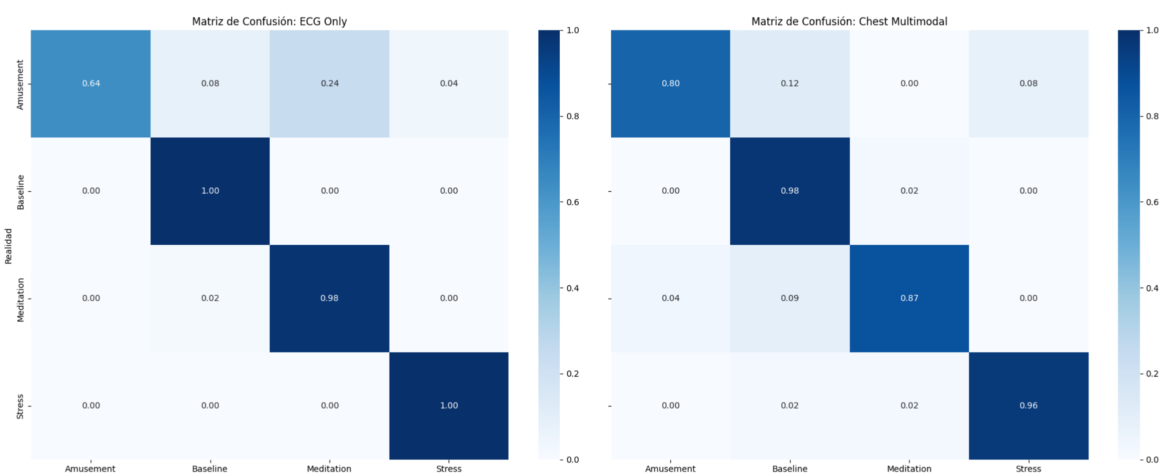

Figure 5: Final Evaluation: The "ECG Only" matrix (Left) demonstrates perfect stress detection capability in the test set, proving that multimodal data (Right) is not strictly necessary for this task.

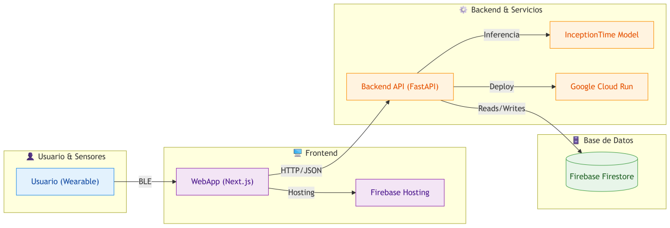

Figure 6: Cloud Architecture: Serverless pipeline.

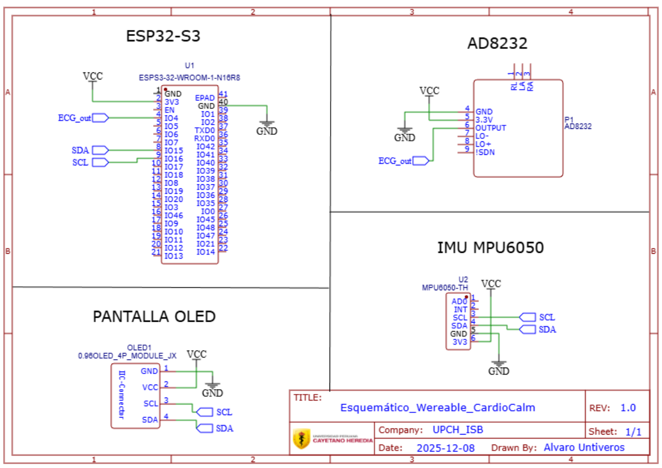

Figure 7: Hardware Schematic (ESP32 + AD8232).

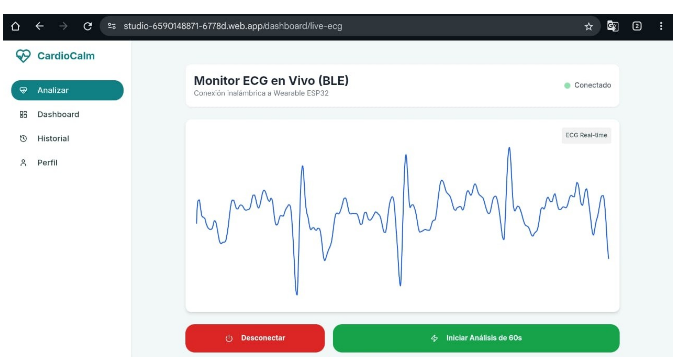

Figure 8: Frontend: Real-time ECG visualization via BLE.

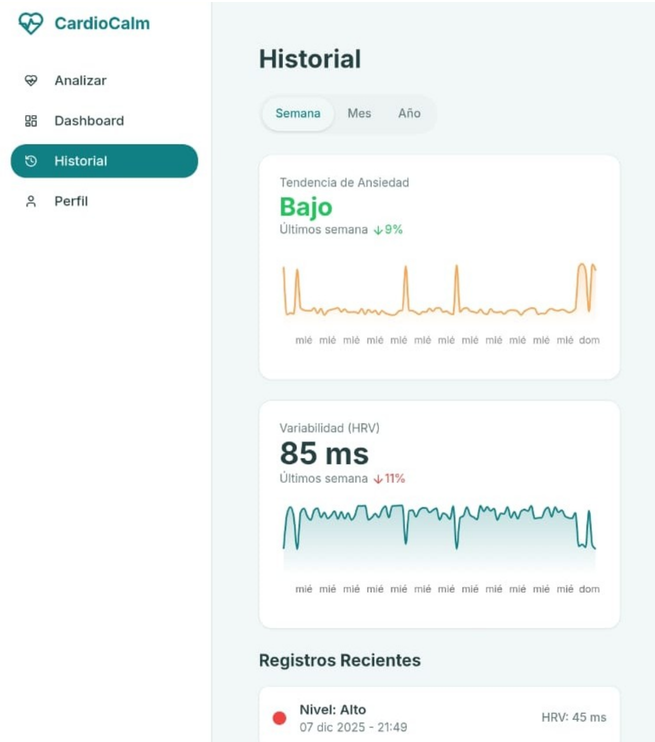

Figure 9: Historical crisis logging screen.

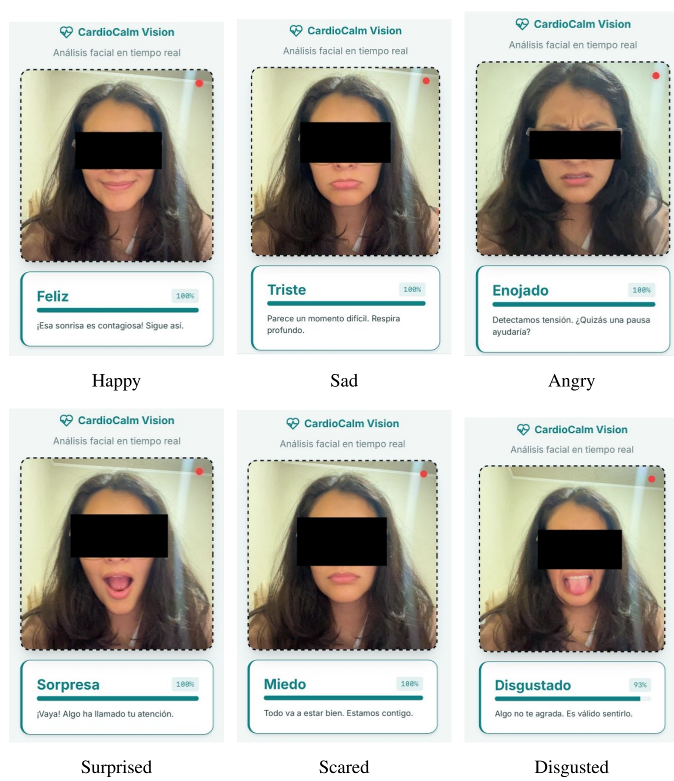

Figure 10: Computer Vision Module: Facial Emotion Recognition.